For all questions on this quiz,

Let

`X = {A, B, C, D, E, F, G, H, I, J}`

Note that

`|X| = 10`

Formulae you may need:

C(n, r) = n! / (r! * (n - r)!)

P(n, r) = n! / (n - r)!

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	Time: 10,100 minutes	Score: 50 out of 100

Score for this quiz: **50** out of 100
Submitted Feb 12 at 6:23pm
This attempt took 10,100 minutes.

Correct answer

Question 1

25 / 25 pts

How many 4 character strings (note - not words) can be formed from the letters of the set X if repetition is allowed?

X = {A, B, C, D, E, F, G, H, I, J}

10,000

5040

210

1

Wrong answer

Question 2

0 / 25 pts

How many length-4 strings can be formed from the letters of X if repetition is not allowed?

X = {A, B, C, D, E, F, G, H, I, J}

10,000

5040

210

1

Wrong answer

Question 3

0 / 25 pts

How many length-4 strings can be formed from the letters of X if repetition is not allowed and the letters must appear in alphabetical order?

X = {A, B, C, D, E, F, G, H, I, J}

10,000

5040

210

1

Correct answer

Question 4

25 / 1 pts

How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

13

9

49

52

Next Quiz: 4

Instructions

For **all** 4 questions on this quiz, we'll be working with the following 2 sets:

A = {1, 3, 5}

B = {1, 2}

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	Time: 5 minutes	Score: 100 out of 100

Score for this quiz: **100** out of 100
Submitted Feb 5 at 6:15pm
This attempt took 5 minutes.

Correct answer

Question 1

25 / 25 pts

Which of the following is A ∪ B?

{3, 5}

{1}

ϕ

{x | x ∈ Z}

{1, 2, 3, 5}

Correct answer

Question 2

25 / 25 pts

Which of the following is A ∩ B?

{3, 5}

{1}

ϕ

{x | x ∈ Z}

{1, 2, 3, 5}

Correct answer

Question 3

25 / 25 pts

What is | A x B |?

Reminder |set| indicates cardinality in this case - not absolute value.

6

1

8

64

Correct answer

Question 4

25 / 25 pts

What is | P(A)|?

Note - in this case, P is my symbol for the powersest - it's the fanciest P canvas will let me have.

6

1

8

64

Next Quiz: 3

Quiz 3

- **Due** No due date **Points** 100 **Questions** 4 **Time Limit** None

Instructions

In this quiz, I'll be using the subscript notation we discussed in class to indicate what base a number is

So if you see 11_{10} , that will indicate the number 11 in base 10

However, you don't need to do the same in your answers. No subscripts are expected in your responses, just the numbers

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	Time: 19 minutes	Score: 100 out of 100

Score for this quiz: **100** out of 100
Submitted Jan 29 at 6:20pm
This attempt took 19 minutes.

Question 1

25 / 25 pts

Convert 49_{10} into base 2 (binary) and explain your work.

Your Answer:
128, 64, 32, 16, 8, 4, 2, 1

00110001

Here i just found the largest number that would go into 49, and then the other numbers that added up to make 49.

so 32 + 16 + 1 which comes out to be 00110001

Question 2

25 / 25 pts

Convert 1011011_2 into base 10 and explain your work

Your Answer:
128, 64, 32, 16, 8, 4, 2, 1

0. 1. 0. 1. 1. 0. 1. 1

so here i take 64 + 16 + 8 + 2 + 1 = 91

Question 3

25 / 25 pts

The following number is in signed binary:

1011011_2

What number does this represent in base 10? Show your work.

Your Answer:
in signed binary the first number 1 would show it's a negative number. So the first one you would use is:

128, 64, 32, 16, 8, 4, 2, 1

1. 0. 1. 1. 0 1 1

so you would add

16 + 8 + 2 + 1 = -27

and the one at the left tells you it's a negative number

Question 4

25 / 25 pts

The following number is in 2's complement form

$1111\ 0110_2$

What number does this represent in decimal? Show your work.

Your Answer:
128, 64, 32, 16, 8, 4, 2, 1

1. 1. 1. 1. 0 1. 1. 0

0. 0. 0. 0. 1. 0. 0 1

+. 1

0 0. 0. 0. 1. 0. 1 0

first you switch the numbers as shown above.

Then add one which becomes 00001010

then convert to decimal. so this becomes -10

Quiz 2:

Question 1

25 / 25 pts

Use a truth table to show that ¬(P → Q) is logically equivalent to P ∧ (¬Q).

P	**Q**	**¬(P → Q)**	**P ∧ (¬Q)**
T	T	F	[Select] ["T", "F"]
T	F	[Select] ["T", "F"]	[Select] ["T", "F"]
F	T	[Select] ["T", "F"]	[Select] ["T", "F"]
F	F	[Select] ["T", "F"]	F

Answer 1:

F

Answer 2:

F

Answer 3:
T

Answer 4:
T

Answer 5:

F

Answer 6:

F

Answer 7:

F

Answer 8:

F

Question 2
12.5 / 25 pts
Translate the following statement into symbolic logic

```
If a number x is a multiple of 6, then x is even
```

To help with this, here are two propositions I've defined for you to use

```
P(a) = a is a multiple of 6
Q(b) = b is even
```

Additionally, here are some symbols we've discussed in class. Some may be helpful

∧
∨
¬

∃
∀
∈

→
↔

Your Answer:
∃x∈P→xQ

You've got the right start but something here isn't quite right!

Question 3

25 / 25 pts

Is the following truth table correct

P	**Q**	**P ↔ Q**
T	T	T
T	F	F
F	T	T
F	F	T

Yes, This is correct

No, this is the truth table for logical AND

No, this is the truth table for $P \rightarrow Q$

No, this is the truth table for logical OR

Question 4

25 / 25 pts

Translate the following symbolic logic into English. Then indicate if the statement is true or false.

Q(a, b) = b is equal to (a * -1)

$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, Q(x,y)$

Reminder: $\mathbb{Z}$ is our shorthand symbol for the set of all integers. Integers are whole numbers (no decimal points) and they can be positive, negative, or 0.

Your Answer:
For all x's in the set of integers, there exists an integer y where b is equal to a times -1.

true because b could be a negative integer and whenever you multiply a by negative 1 it will also become negative.

a = 2

b= -2

-2 = 2 x -1

or

a = 3

b= -3

-3 = 3 x -1

Quiz Score: **87.5** out of 100

Quiz 1:

Instructions

For the following problems you may find you need some of our laws of equivalence:

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	Time: 25 minutes	Score: 94.75 out of 100

Score for this quiz: **94.75** out of 100
Submitted Jan 15 at 6:28pm
This attempt took 25 minutes.

Correct answer

Question 1

25 / 25 pts

Complete the following truth table

P	Q	R	¬(P ∨ (Q ∨ R))	(¬ P) ∧ (¬ Q) ∧ (¬ R)
F	F	F	[Select] ["T", "F"]	[Select] ["T", "F"]
F	F	T	F	F
F	T	F	F	[Select] ["T", "F"]
F	T	T	[Select] ["T", "F"]	[Select] ["T", "F"]
T	F	F	[Select] ["T", "F"]	[Select] ["T", "F"]
T	F	T	F	[Select] ["T", "F"]
T	T	F	[Select] ["T", "F"]	[Select] ["T", "F"]
T	T	T	[Select] ["T", "F"]	[Select] ["T", "F"]

Answer 1:

T

Answer 2:

T

Answer 3:

F

Answer 4:

F

Answer 5:

F

Answer 6:

F

Answer 7:

F

Answer 8:

F

Answer 9:

F

Answer 10:

F

Answer 11:

F

Answer 12:

F

Answer 13:

F

Answer 14:

F

Answer 15:

F

Answer 16:

F

Correct answer

Question 2

25 / 25 pts

Based on the above truth table, are **¬(P ∨ (Q ∨ R))** and **(¬ P) ∧ (¬ Q) ∧ (¬ R)** logically equivalent?

Yes - DeMorgan's law shows this

Yes- the Distributive law shows this

Yes - the Associative law shows this

No - DeMorgan's law does not work that way

No - the Distributive law does not work that way

No - the Associative law does not work that way

Question 3

21 / 25 pts

Use our list of logical equivalence laws to demonstrate why **P ∧ Q** and **¬ (¬ P ∨ ¬Q)** are logically equivalent

Your Answer:

using DeMorgan's Law becomes:

¬(P ∧ Q)

¬(¬P ∨ ¬Q) Law of double negatives

P ∨ Q

I'm not showing that they're logically equivalent.

P	Q	P∧Q	¬P	¬P ∨ ¬Q	¬(¬P ∨ ¬Q)
T	T.	T.	F.	F	T
T	F.	F.	F.	T	F
F	T	F.	T.	T	T
F	F.	F.	T.	T	T

Question 4

23.75 / 25 pts

For the following fill in the blank answers, indicate whether they are **propositions**

A) Every real number is an even integer. Proposition

B) 7 is greater than 4 Proposition

C) Later, we can go to the store Not Proposition

D) What time is it Not Proposition

E) X < 12 Proposition

Answer 1:

Proposition

Answer 2:

Proposition

Answer 3:

Not Proposition

Answer 4:

Not Proposition

Answer 5:

Proposition

Not Proposition